

MASTER OF COMPUTER**APPLICATIONS/BACHELOR OF****COMPUTER APPLICATIONS****(REVISED) (MCABCA)****Term-End Examination****June, 2023****MCS-012 : COMPUTER ORGANIZATION AND
ASSEMBLY LANGUAGE PROGRAMMING***Time : 3 Hours**Maximum Marks : 100**(Weightage : 75%)***Note : Question No. 1 is compulsory and carries****40 marks. Attempt any three questions from
the rest.**

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1. (a) Describe the structure of 8086 micro-processor with the help of a diagram. 6
 - (b) What is an instruction cycle ? Explain with the help of a flowchart. 6

P. T. O.

- (c) Represent the following numbers using IEEE-754 floating point single precision number format : 4
 - (i) 1010.0001
 - (ii) - 0.0000111
- (d) Explain instruction pipeline with the help of a diagram. 4
- (e) What are the different kinds of Interrupts ? How does CPU know that an interrupt has occurred ? 5
- (f) What is DMA ? Explain the functions of DMA. 5
- (g) Consider a four variable Boolean function :

$$F = \Sigma (0, 4, 6, 7, 8, 10, 11, 15)$$
 Minimize this function using K-map and draw the resultant function using logic gates. 6
- (h) Convert decimal number $(49.25)_{10}$ into : 4
 - (i) binary
 - (ii) hexadecimal
 - (iii) octal
2. (a) Explain the differences between micro-programmed control and hardwired control. 5

- (b) Draw and explain the logic diagram of a 3×8 decoder. 5
- (c) Design a combinational circuit with three inputs x, y, z and three outputs A, B, C . When the binary equivalent of input to this circuit is 0, 1, 2 or 3 the binary output is one greater than the input. When the binary equivalent of input is 4, 5, 6 or 7 the binary output is one less than the input. Also draw the truth table. 6
- (d) Find the 9's and 10's complement of 128. Then convert 128 to binary and find 1's and 2's complement. 4
3. (a) Add 25 and (-25) in binary using 8-bit register for the following representations : 6
- Signed magnitude representation
 - Signed 1's complement
 - Signed 2's complement
- (b) Explain the following instructions of 8086 assembly language with the help of an example : 8
- ADC
 - MUL
 - XOR
 - ROL

P. T. O.

- (c) Explain the role of Interrupt Vector Table (IVT) in the context of 8086 micro-processor with the help of a diagram. 6
4. (a) The following memory units are specified by the number of words times the number of bits per word. How many address lines and input-output data lines are needed in each case ? 4
- $2K \times 16$
 - $64K \times 8$
- (b) What is a micro-operation ? Explain the sequence of micro-operations required to fetch an instruction stored in the memory in the context of a Von Neumann machine. 6
- (c) Explain the following addressing modes with the help of an example of each : 6
- Register addressing
 - Index addressing
 - Direct addressing
- (d) Find the even and odd parity bits for the following 7-bit data : 4
- 1010101
 - 0000111
5. (a) Explain the terms ASCII and UNICODE. 3

- (b) Register 'A' holds 8-bit binary 11011001. Determine B operand and the logic micro-operation to be performed in order to change the value in A to :
- 6
- (i) 0000 1001
- (ii) 1111 1001
- (iii) 0010 0110
- (c) Write the assembly language code using 8086 assembly language for performing the following operation :
- 6
- $$Z = ((A - B) / 10 * C) ** 2$$
- (d) Explain the use of circular overlapped register window in a RISC processors with the help of a diagram.
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**MASTER OF COMPUTER
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COMPUTER APPLICATION
(REVISED) (MCA/BCA)
Term-End Examination**

June, 2024

**MCS-012 : COMPUTER ORGANISATION AND
ASSEMBLY LANGUAGE PROGRAMMING**

Time : 3 Hours

Maximum Marks : 100

Note : *Question No. 1 is compulsory and carries
40 marks. Attempt any **three** questions from
the rest.*

1. (a) What are the components of an EXE
program ? What are the advantages of
using EXE programs ? 5

- (b) Explain the Instruction Pipelining with the help of a diagram. 5
- (c) What is full-adder ? Draw the logic diagram of a full-adder circuit. 6
- (d) How is the access time calculated on a hard disk ? 4
- (e) Simplify the following Boolean function using K-map : 6

$$F(A, B, C, D) = \Sigma (0, 4, 5, 6, 12, 13, 14)$$

Also draw the logic diagram of the simplified circuit using AND, OR and NOT gates.

- (f) What is the need of Cache memory ? Explain the direct mapping cache organisation with the help of a diagram/example. 6

- (g) What is the sequence of micro-operations required for fetching an instruction from the memory. You may assume that machine uses Accumulator (AC) register, Program Counter (PC) register, Instruction Register (IR), Memory Address Register (MAR) and Data Register (DR). 5
- (h) What is the role of Int 21h in 8086 micro-processor ? 3
2. (a) What is an interrupt ? What happens on the occurrence of an Interrupt ? Explain with the help of a diagram. 6
- (b) Explain Von-Neumann architecture with the help of a diagram. 7

- (c) Draw and explain the logic diagram of J-K flip-flop. Also, make the characteristic table of J-K flip-flop. 7
3. (a) Find 1's and 2's complement of the following fixed point numbers : 4
- (i) 10100010
- (ii) 00000000
- (b) Represent the following numbers in floating point single precision number format : 6
- (i) 1010.0001
- (ii) -0.0000111
- (c) Explain the following addressing modes with the help of an example of each : 10
- (i) Immediate addressing
- (ii) Direct addressing

- (iii) Register addressing
 - (iv) Indirect addressing
 - (v) Indexed addressing
4. (a) Explain the programmed I/O technique with the help of a flow chart. How is it different from DMA ? 8
- (b) Explain the following terms in the context of Input/Output system : 12
- (i) Device driver
 - (ii) Refresh rate of a monitor
 - (iii) Classification of printers
 - (iv) Modem
5. (a) Write a program using 8086 assembly language that converts an ASCII digit to equivalent binary value. 6

- (b) What is the need of segment registers in 8086 micro-processor ? Explain the use of code segment and stack segment registers with the help of an example of each. 8
- (c) Explain the following 8086 assembly commands : 6
- (i) XCHG instruction
 - (ii) SHL/SHR instruction

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MCS-012

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June, 2020

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ASSEMBLY LANGUAGE PROGRAMMING**

Time : 3 Hours

Maximum Marks : 100

Weighthage : 75%

*Note : (i) Question No. 1 is compulsory and carries
40 marks.*

*(ii) Attempt any three questions from the
rest.*

1. (a) Convert the following numbers as stated : 6

(i) $(23.125)_{10}$ to binary

(ii) $(36.5)_{10}$ to octal

(iii) $(135)_{10}$ to hexadecimal

(b) Draw the truth table for the following Boolean function :

$$F = (A \cdot B + C) + (\bar{A} \cdot C) + (\bar{B} \cdot A \cdot \bar{C})$$

Use k-map to simplify the above Boolean function. 5

(c) What is the need of Cache memory ?

Explain the direct Cache mapping scheme with the help of an example/diagram. 6

(d) Explain the interrupt-driven I/O technique with the help of a diagram. 4

(e) How is the next instruction that is to be executed brought into Instruction Register for execution ? Explain the sequence of micro-operation that are needed to perform this operation. Which of these micro-operations will take longest time to execute ? Give justification in support of your answer. 6

(f) How is the large register file of RISC useful ? 3

(g) Write a program using 8086 assembly language that finds the larger of two byte values stored in two memory locations named A and B respectively. The larger of two values should be stored in AL register. 6

- (h) Calculate the physical address for the following segment register : 4

Offset (or Register) pair

- (i) Offset of data byte in segment $(0200)_h$

Data Segment (DS) : $(1FFF)_h$

- (ii) Code Segment Register (CS) : $0F10$

Instruction Pointer Register (IP) :

2562

2. (a) Explain the 'Stored Program Concept' for a Von Neumann machine. 4

- (b) What is an I/O processor ? Explain its characteristics. Explain the selector and multiplexer channels with the help of diagram(s). 6

- (c) Consider the Registers R1 having value $(1011\ 0101)_2$ and R2 having value

$(0110\ 0111)_2$. Perform the following operations using R1 and/or R2. The result should be stored in a register R :

(i) Addition of R1 and R2 with carry

(ii) Decrement R1

(iii) Increment R1

(iv) Subtract R2 from R1

It may be noted that only addition micro-operation is allowed. 6

(d) Explain the FAR procedure call in the context of 8086 assembly language. 4

3. (a) What is an Interrupt Vector Table in 8086 microprocessor ? How is it used to process an Interrupt ? Explain with the help of a diagram. 6

- (b) Explain the following in the context of micro-programmed control unit : 6
- (i) Control memory
 - (ii) Sequencing logic
 - (iii) Vertical micro-instruction
- (c) Explain the following in the context of printing technology : 6
- (i) Print quality
 - (ii) Impact and non-impact printers
 - (iii) Print resolution
- (d) List any *four* advantages of densely packed integrated circuits. 2
4. (a) What is the use of Multiplexers ? Draw and explain the logic diagram of a 4×1 multiplexer. Also draw the truth table for this multiplexer. 6

(b) A memory chip has a capacity of $1 \text{ M} \times 16$ bits : 4

(i) How many address lines does it have ?

(ii) What is the capacity of the chip in bytes ?

(c) What is an Accumulator base Instruction Set Architecture ? Write the assembly code for the expression $A = B * C + D$ for Accumulator based machine. 6

(d) What is the role of Flag register in 8086 microprocessor ? Explain the role of any *three* flags in this register. 4

5. Explain briefly any *eight* of the following :

$$8 \times 2\frac{1}{2} = 20$$

(a) Assembler

(b) Stack segment

- (c) EXE programs
- (d) Shift instruction
- (e) D flip-flop
- (f) Memory interleaving
- (g) Latency time in disk access
- (h) Normalization of floating point numbers
- (i) Unicode
- (j) Counters

MCA (Revised) / BCA (Revised)**Term-End Examination****June, 2022****MCS-012 : COMPUTER ORGANISATION AND
ASSEMBLY LANGUAGE PROGRAMMING***Time : 3 hours**Maximum Marks : 100**(Weightage : 75%)*

Note : Question number 1 is **compulsory** and carries 40 marks. Attempt any **three** questions from the rest.

1. (a) Simplify the following Boolean function in SOP form by using K-map. Also draw the logic diagram of the simplified function using AND-OR-NOT gates : 5

$$F(A, B, C, D) = \Sigma (0, 2, 4, 6, 9, 11, 14)$$

- (b) Draw the logic diagram of D flip-flop. Explain its working. Make the characteristics table and excitation table for D flip-flop. 5
- (c) Why is DMA needed ? What are the different functions that a DMA interface should perform ? 3

- (d) A disk rotates at a speed of 6000 rpm (revolutions per minute). It has a seek time of 10 milliseconds. The disk has 100 tracks with each track having 200 sectors. Find the average access time of this disk. 3
- (e) A RAM chip has a capacity of $16\text{ K} \times 4$ bits. Answer the following questions for this RAM : $1+1+2=4$
- How many numbers of input and output lines does this memory have ?
 - How many address lines does it have ?
 - How many such chips will be needed to construct a RAM of size 64 K Byte ?
- (f) Perform the following micro-operations in a sequence to registers R1 and R2. The initial value of R1 is 00111111 and R2 is 11000001. 5
- $R \leftarrow R1 + R2 + 1$ (Add with carry)
 - $R1 \leftarrow R2 + (\text{All 1s})$
in a Register (Decrement)
 - $R2 \leftarrow R1$ (Transfer R1)
 - $R \leftarrow R1 \wedge R2$ (AND)
 - $R \leftarrow \overline{R1}$ (Complement R1)
- (g) Explain the concept of Micro-program Control Unit. What is the need of control memory ? How is control memory organised ? Explain with the help of a diagram. 5

- (h) Consider the following pair of registers or offsets of size 16 bits of 8086 microprocessor. Explain how these registers or offset pairs are used to compute physical memory address of 20 bits. Show actual address computation : 4
- (i) DS has $(25CD)_h$ and offset $(004A)_h$
- (ii) SS has $(2F1A)_h$ and SP has $(1124)_h$
- (i) Write a program using 8086 assembly language that finds the smallest of three byte values stored in the memory. 6
- 2.** (a) What is a multiplexer ? Draw the truth table and logic diagram of 2×1 multiplexer and explain its working. 4
- (b) What is half adder ? Draw the truth table for a half adder and construct the logic diagram of a half adder using AND, OR and NOT gates only. 4
- (c) Explain how an even parity bit can identify error using an example of 4 bit data. Can it identify the bit which is in error ? Give reason in support of your answer. 4
- (d) Perform the following operations using signed 2's complement notation of 8 bits. Also indicate overflow, if any. 4
- (i) $+59 + 83$
- (ii) $+59 - 83$
- (iii) $-59 + 69$
- (iv) $-59 - 69$
- (e) Explain the concept of Von Neumann machine, with the help of a diagram. 4

3. (a) Draw the logic diagram of a RAM cell using a J-K flip-flop. Explain various input and selection lines for this cell. Explain how reading or writing can be performed on this cell. 7
- (b) Explain the direct cache mapping scheme for a memory of size 4 K Byte and cache of size 128 byte. The main memory is byte addressable and size of one slot/line of cache is also 1 byte. 5
- (c) Explain the characteristics of the following I/O devices/interfaces : 6
- (i) DVD-ROM
- (ii) LCD monitors
- (iii) Scanner
- (d) Define the term “Interrupt” in the context of a computer, with the help of an example. 2
4. (a) Assuming that a machine has instructions of size 16 bits only. How many bits of this instruction will be required for the following ? $1+1+2=4$
- (i) Length of operation code (op-code) if the machine has 50 instructions.
- (ii) Length of addressing mode bits if machine has four different addressing modes.
- (iii) If one of the four addressing modes is direct addressing, then what would be the size of main memory address that can be supported by this machine ?

- (b) What is the role of memory stack for implementation of subroutine call and return instructions ? Explain with the help of suitable diagrams. 6
- (c) Explain the functioning of Wilkes control unit, with the help of a diagram. 5
- (d) How is a large register file used in RISC machine ? Explain with the help of a diagram. 5
- 5.** (a) Explain the following instructions of 8086 microprocessor, with the help of an example each : 8
- (i) XCHG
- (ii) LEA
- (iii) ADC
- (iv) XOR
- (b) Explain the differences between ROL and RCL instructions of 8086 microprocessor, with the help of an example each. 4
- (c) Explain the role of Interrupt Vector Table (IVT), for processing of an Interrupt in 8086 microprocessor, with the help of a diagram. 5
- (d) What is an Assembler ? Explain its purpose. 3
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Note : *Question number 1 is **compulsory** and carries 40 marks. Attempt any **three** questions from the rest.*

1. (a) Perform the following computations using *signed 1's complement* notation of length 8 bits. Also indicate overflow, if any : 6
- (i) $-76 - 52$
- (ii) $+79 + 49$
- (iii) $+79 - 86$
- (b) Design a full-adder circuit using K-map. 4
- (c) Explain the two-way set associative cache mapping scheme with the help of an example. 5
- (d) What is DMA ? Why is it useful ? Draw the block diagram of a DMA interface. 5

- (e) Instructions of machine are such that they have two register operands. However, to load a register a special instruction has been designed which either contains the operand value or address of the operand. List and explain four addressing modes for this machine. 4
- (f) What is the role of control memory in a micro-programmed control unit ? Explain the organisation of control memory with the help of a diagram. What is a horizontal micro-instruction ? Explain. 6
- (g) Write a program using 8086 assembly language that moves content of byte memory location X1 and X2 to AL and BL registers. The program then finds the larger value of AL or BL register and stores it in DL register. 6
- (h) Assume the following values in the registers :
Instruction Pointer (IP) contains $(A521)_h$
Stack Pointer (SP) contains $(00FF)_h$
Code Segment (CS) contains $(0FFF)_h$
Stack Segment (SS) contains $(000F)_h$
Find the following using the above information : 4
- (i) Physical address of top of stack
 - (ii) Physical address of instruction

- 2.** (a) Draw logic diagram to implement AND, OR and NOT operations using NAND gate(s). 5
- (b) Explain the following in the context of floating point number representation with the help of an example : 5
- (i) Normalised mantissa
- (ii) Biased exponent
- (c) How many RAM chips of size $512\text{ K} \times 1\text{ bit}$ are required to build 1 MB memory ? 2
- (d) What is Programmed Input/Output ? Explain with the help of a diagram. Explain the difference between Programmed I/O and Interrupt driven I/O. 6
- (e) What is Latency time in the hard disk ? 2
- 3.** (a) Explain the steps required to fetch an instruction from a memory location to instruction register with the help of micro-operations. 5
- (b) What will be the length of various fields of an instruction considering the following ? 5
- (i) 64 possible operations
- (ii) 8 addressing modes
- (iii) Memory size of 4 KB (byte addressing is used)
- (iv) It has 32 registers
- (v) Each instruction has one register and one memory operand
- Make suitable assumptions.

- (c) Explain the concept of NEAR and FAR procedural calls in 8086 microprocessor with the help of one example each. 6
- (d) Explain the use of INT 21h in 8086 microprocessor for reading a single character from the keyboard with the help of an example. 4
4. (a) Draw and explain the truth table and logic diagram of a 3-bit synchronous counter. 5
- (b) Explain the von Neumann architecture with the help of a diagram. 5
- (c) What is an Input/Output processor ? How is it different from DMA ? 4
- (d) Differentiate between the following : 6
- (i) SRAM and DRAM
- (ii) ROM and Flash Memory
5. (a) What is the use of stack in subroutine CALL instruction ? Explain using an example. 5
- (b) Why is RAID used in computers ? What is RAID Level 0 ? 3
- (c) Explain the following assembly language instructions with the help of an example each : 8
- (i) MUL
- (ii) ADD
- (iii) TEST
- (iv) SHR
- (d) Explain the use of CX register in implementing looping in 8086 assembly language. 4